

Solution Requirements

Date	30 September 2026
Team ID	SWTID-2026-7291
Project Name	Weather Wise
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can access weather information without logging in or creating an account.	High
2	Authorization levels	All users can search for cities and view current weather conditions and forecasts	Median

3	External interfaces	The system connects to a Weather API to retrieve real-time temperature, humidity, wind speed, and rainfall information.	High
4	Transactions processing	The system sends requests to the Weather API, processes the received data, and displays the results to users.	High
5	Reporting	The system displays current weather conditions and forecast information through a user-friendly dashboard.	High
6	Business rules, etc.	Users can search for a city and view its weather details. The system displays an appropriate message if the city is invalid or data is unavailable.	High
7	Compliance to laws or regulations	Compliance to Laws or Regulations The system respects user privacy and handles location information responsibly.	Median
8	<i>Other</i>	The system provides temperature, humidity, wind speed, rainfall, and weather conditions through a simple interface.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should retrieve and display weather information quickly.	Weather data should load within 3 seconds under normal network conditions.
2	Scalability	The system should support multiple users searching for weather information simultaneously.	Supports at least 100 concurrent users under normal operating conditions.
3	Security & Data Privacy	The system should protect API credentials and avoid unnecessary collection of personal information.	API keys must be stored securely and not exposed in the frontend.
4	Reliability & Availability	The system should provide weather information consistently and handle API failures gracefully.	Displays an appropriate error message when the weather service is unavailable
5	Usability & Accessibility	The application should have a simple, responsive, and user-friendly interface.	Users should be able to search for a city and view weather information easily on desktop and mobile devices.
6	Maintainability & Portability	The application should be easy to maintain and accessible across different devices and browsers.	The system should work on modern web browsers and allow future feature enhancements.

